

Kuldeep Lodhi

SENIOR SOFTWARE ENGINEER

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Summary

Senior Software Engineer with 7+ years architecting data-intensive platforms across logistics, retail, inventory operations, **AI platforms**, and **algorithmic trading**. Specialized in React, TypeScript, UI/UX-focused frontend architecture, **backend architecture**, **API signatures and integration**, and real-time systems (Kafka, SSE). Proven **Team Lead** shipping enterprise platforms serving 1,900+ stores, 12,000+ daily users, and INR 1,000 Cr in managed inventory.

Professional Experience

Reliance Jio Platforms

Jun 2024 -- Present

Software Development Engineer – 2

Navi Mumbai, India

- Architected the frontend platform layer of an enterprise inventory system that gives 12,000+ daily users live control over INR 1,000 Cr of inventory across 1,900+ stores and 6 retail brands
- Designed **REST API contracts and request/response signatures** with backend engineers and authored **stored procedures** for high-volume stock aggregations, removing integration rework and holding dashboard response times steady
- Drove **system architecture brainstorming and design reviews** for new modules, converting business requirements into scalable service boundaries and event contracts adopted team-wide
- Established a shared library of 50+ reusable TypeScript components and a standardized Redux architecture, cutting duplication by 30%, and re-engineered dashboard rendering with memoization and virtualization for a 45% gain
- Led a 4-member frontend team as **Team Lead**, instituting UI/UX standards and 30+ code reviews per sprint, and introduced Docker and Azure DevOps CI/CD pipelines for frequent, low-risk releases

Technologies: React, TypeScript, Redux, Node.js, REST APIs, Kafka, PostgreSQL, Docker, Azure DevOps

Mobile Programming (Client: Reliance Jio)

Dec 2023 -- Jun 2024

Software Developer

Mumbai, India

- Embedded with Jio's internal engineering team to build the foundation of the enterprise inventory management platform, establishing the React and TypeScript baseline the product still runs on
- Aligned every module to existing architecture and coding standards, keeping the codebase maintainable as the platform and team scaled
- Partnered with backend engineers on **API integration and payload design** for high-volume stock data, reducing over-fetching and simplifying client-side state management

Algonauts Advisory

Nov 2021 -- Dec 2023

Senior Software Developer

Mumbai, India

- Architected a full-stack **algorithmic trading simulator** that scaled to 400 daily users and 8K trades/day, establishing the company's core product offering
- Designed backend APIs and data models for order execution, portfolio state, and market-data ingestion, **brainstorming architecture directly with founders** to convert trading workflows into a data model that absorbed growing trade volume as the platform scaled
- Delivered an ML-based portfolio recommendation engine that increased user engagement by 25%
- Scaled the engineering team from 1 to 3 developers while sustaining 99.5% uptime through coding standards, reviews, and onboarding documentation

Technologies: Svelte.js, Node.js, Python, REST APIs, PostgreSQL, Machine Learning

Ugam Solutions

Jun 2019 -- Nov 2021

Senior Analyst

Thane, India

- Shipped 15 enterprise analytics projects for CDC, University of Chicago, Google Screenwise, and IPSOS, supporting client research decisions with reliable data products
- Built reusable JavaScript libraries, dashboards, and **SQL stored procedures** for survey data transformation, cutting report generation time by 40% and improving data-refresh reliability across recurring deliverables

Key Projects

AnyTrac RFID Platform

React, TypeScript, Redux, Node.js, Kafka, Google Maps SDK

- Rolled out unified RFID-based inventory tracking that replaced 5-minute manual counts with 50-item carton scans in 1 second, transforming store-level stock-taking throughput
- Designed an **event-driven architecture** publishing Product **Stock-on-Hand (SOH)** and tracking updates to Kafka queues, backed by **API signatures and stored procedures** for high-volume SOH aggregation that held portal response times stable as the platform expanded
- Powered real-time GPS tracking, route visualization, and floor-plan mapping through **Google Maps SDK**, giving operations zone-level stock visibility and in-transit exception handling
- Centralized marketing and analytics tag deployment through a **Tag Management System (TMS)**, eliminating manual code changes for tracking updates across store portals

Impact: Raised inventory accuracy from 75% to 98%, reduced shrinkage by 40%, and powered Store, Cluster, and Self-Checkout portals across 1,900+ retail locations

JVA Retail Video Analytics Platform

React, TypeScript, Konva, Kafka, Real-time Streaming

- Built the dashboard layer for an **AI-powered video analytics platform** turning live camera streams into retail and safety insights, spanning traffic analytics, queue monitoring, demographics, and heatmap modules
- Extended the platform into **security surveillance and safety monitoring**, adding **fire detection** and intrusion/anomaly analysis on live CCTV feeds to surface incidents to store teams within seconds
- Shipped a store planner for floor-plan mapping and camera placement, enabling zone-based visitor analytics
- Integrated a Kafka streaming pipeline fanning camera feeds out to multiple analytics consumers evaluating defined Regions of Interest (ROIs), allowing new analytics pipelines to be added without touching ingestion
- Implemented **Server-Sent Events (SSE)** for real-time alerting on fire and security events, queue thresholds, and congestion, moving store teams from reactive to proactive response

Reliance Jewels Inventory Platform

React, TypeScript, Redux, Node.js

- Consolidated daily reconciliation, SOH, and inward/outward movement into a single system of record for Reliance Jewels stores, backed by reporting APIs that cut reconciliation time for store managers

Reliance Foundation Scholarship Portal

React, Node.js, MongoDB

- Scaled a scholarship management system to 50,000 annual applications with OCR-based document ingestion, removing the manual verification bottleneck
- Migrated a legacy K2 workflow system to a **MERN architecture**, improving scalability by 200% and cutting dependency on licensed workflow tooling
- Built REST APIs and MongoDB schemas for application, evaluation, and interview workflows, supporting 50 concurrent interview sessions with automated evaluation

Technical Skills

Frontend & UI/UX: React, TypeScript, Next.js, Redux, Svelte, Konva, Chart.js, ApexCharts, Material-UI, Tailwind CSS

Backend & APIs: Node.js, Express, Python, Django, REST API Design, API Signatures & Contracts, Stored Procedures, Microservices, Authentication & Authorization

Architecture & Streaming: Scalable & Distributed Systems, Event-driven Architecture, Service Decomposition, Design Reviews, Apache Kafka, Server-Sent Events (SSE), Tag Management System (TMS)

Maps & Geospatial: Google Maps SDK/API, GPS Tracking, Route Visualization, Geofencing, Floor-plan/Facility Mapping

AI & ML Integration: AI Video Analytics, Fire & Security Detection, ML-based Recommendation Systems

Database: PostgreSQL, MongoDB, MySQL, Redis, Query Optimization

DevOps, Cloud & Performance: Docker, CI/CD, Azure DevOps, AWS, Git, Memoization, Bundle Optimization

Leadership & Collaboration: Agile/Scrum, Team Lead (4 Engineers), Code Reviews (30/Sprint), Stakeholder Management

Education

Bachelor of Engineering in Computer Engineering

Vidyalankar Institute of Technology, Mumbai

2015 – 2019

CGPI: 7.95/10